

# PROS9-A950-OSB-M

Serie PROS-M · R134A · senza serbatoio / without tank



PROS9-A950-OSB-M

RESA FRIGORIFERA · COOLING CAPACITY

621 kW

EER · 12/7 °C · AIR 35 °C

2,99

RESA NOMINALE · 15 °C · AIR 25 °C

944 kW

PORTATA ACQUA · WATER FLOW

106,8 m<sup>3</sup>/h

|                            |                   |                                        |                       |
|----------------------------|-------------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 2 × vite          | L × P × H (mm)                         | 8.120 × 2.197 × 2.442 |
| Circuiti / Circuits        | 2                 | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 14                | Taglia / Size                          | 950                   |
| Refrigerante / Refrigerant | R134A · GWP 1.430 | Campo aria esterna / Ambient air range | -5 / +50 °C           |

## Resa in sintesi / Capacity at a glance kW

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 692   | 657   | 621   | 584   | 546   | 508   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 944   | 898   | 851   | 803   | 754   | 703   |

Refrigeratori condensati ad aria con compressori a vite e condensatore a microcanali, selezionabili con acqua in uscita fino a 17 °C: indicati per i processi che lavorano con acqua non troppo fredda.

Air-cooled chillers with screw compressors and micro-channel condenser, selectable with leaving water up to 17 °C: suited to processes that run on water that is not too cold.

# Dati tecnici

## Technical data

PROS9 - A950 - OSB - M

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 944,03 kW                |
| Resa frigorifera / Cooling capacity | 811.866 kcal/h           |
| Potenza assorbita / Power input     | 176,58 kW                |
| EER / EER                           | 5,35                     |
| Portata acqua / Water flow          | 162,37 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 621,03 kW                |
| Potenza assorbita / Power input     | 207,46 kW                |
| EER / EER                           | 2,99                     |
| ESEER / ESEER                       | 3,94                     |
| Portata acqua / Water flow          | 106,82 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                        |
|---------------------------------------------|------------------------|
| Compressori / Compressors                   | 2 × Vite               |
| Circuiti frigoriferi / Refrigerant circuits | 2                      |
| Ventilatori / Fans                          | 14                     |
| Condensatore / Condenser                    | a microcanali          |
| Evaporatore / Evaporator                    | a fascio tubiero       |
| Refrigerante / Refrigerant                  | R134A · GWP 1.430 · A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 8.120 × 2.197 × 2.442 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 · 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 608,6 kW     |
| MEG 30 % | × 0,950 | 590,0 kW     |
| MEG 40 % | × 0,920 | 571,3 kW     |
| MPG 20 % | × 0,975 | 605,5 kW     |
| MPG 30 % | × 0,960 | 596,2 kW     |
| MPG 40 % | × 0,950 | 590,0 kW     |

# Prestazioni Performance

PROS9-A950-OSB-M · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30  | 35  | 40  | 45  | 50  |
|---------------|-------|-------|-------|-----|-----|-----|-----|-----|
| 1             | 594   | 568   | 540   | 512 | 482 | 452 | 422 | 391 |
| 2             | 620   | 592   | 564   | 534 | 504 | 473 | 441 | 409 |
| 3             | 646   | 617   | 588   | 557 | 526 | 493 | 461 | 427 |
| 4             | 673   | 644   | 613   | 581 | 549 | 515 | 481 | 447 |
| 5             | 701   | 670   | 638   | 605 | 572 | 537 | 502 | 466 |
| 6             | 730   | 698   | 665   | 631 | 596 | 561 | 524 | 487 |
| 7             | 759   | 726   | 692   | 657 | 621 | 584 | 546 | 508 |
| 8             | 790   | 756   | 721   | 685 | 647 | 609 | 570 | 530 |
| 9             | 821   | 786   | 750   | 712 | 673 | 634 | 593 | 552 |
| 10            | 854   | 818   | 780   | 741 | 701 | 660 | 618 | 576 |
| 11            | 887   | 850   | 811   | 771 | 729 | 687 | 643 | 599 |
| 12            | 922   | 883   | 843   | 802 | 759 | 715 | 670 | 625 |
| 13            | 957   | 917   | 875   | 833 | 788 | 743 | 697 | 650 |
| 14            | 994   | 953   | 910   | 865 | 820 | 773 | 725 | 677 |
| 15            | 1.031 | 988   | 944   | 898 | 851 | 803 | 754 | 703 |
| 16            | 1.070 | 1.026 | 980   | 933 | 885 | 835 | 784 | 732 |
| 17            | 1.110 | 1.064 | 1.017 | 968 | 918 | 866 | 814 | 760 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 141 | 155 | 169 | 186 | 204 | 224 | 246 | 270 |
| 2             | 142 | 155 | 170 | 186 | 204 | 225 | 247 | 271 |
| 3             | 142 | 156 | 170 | 187 | 205 | 225 | 247 | 272 |
| 4             | 143 | 156 | 171 | 187 | 206 | 226 | 248 | 273 |
| 5             | 143 | 157 | 171 | 188 | 206 | 227 | 249 | 274 |
| 6             | 144 | 157 | 172 | 189 | 207 | 227 | 250 | 274 |
| 7             | 144 | 158 | 172 | 189 | 207 | 228 | 250 | 275 |
| 8             | 145 | 158 | 173 | 190 | 208 | 228 | 251 | 276 |
| 9             | 145 | 159 | 173 | 190 | 209 | 229 | 252 | 277 |
| 10            | 146 | 159 | 174 | 191 | 209 | 230 | 252 | 277 |
| 11            | 146 | 160 | 174 | 191 | 210 | 230 | 253 | 278 |
| 12            | 147 | 160 | 175 | 192 | 210 | 231 | 253 | 279 |
| 13            | 147 | 161 | 175 | 192 | 211 | 231 | 254 | 279 |
| 14            | 148 | 161 | 176 | 193 | 211 | 232 | 255 | 280 |
| 15            | 149 | 162 | 177 | 193 | 212 | 232 | 255 | 280 |
| 16            | 149 | 162 | 177 | 194 | 212 | 233 | 256 | 281 |
| 17            | 150 | 163 | 178 | 194 | 213 | 233 | 256 | 282 |

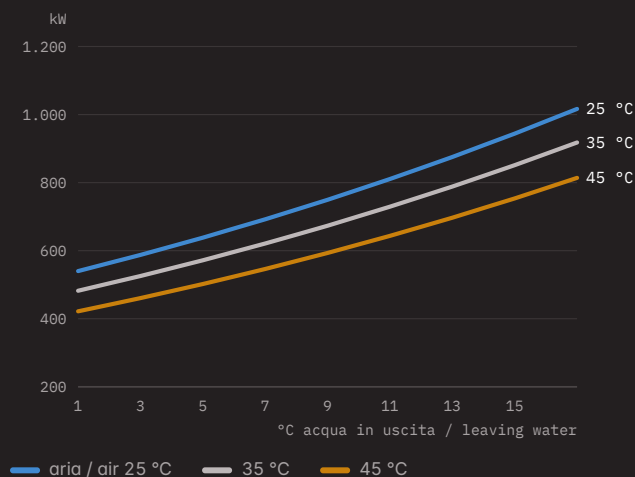
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|---------------|------|------|------|------|------|------|------|------|
| 1             | 4,21 | 3,67 | 3,19 | 2,75 | 2,37 | 2,02 | 1,72 | 1,45 |
| 2             | 4,38 | 3,82 | 3,32 | 2,87 | 2,46 | 2,11 | 1,79 | 1,51 |
| 3             | 4,54 | 3,97 | 3,45 | 2,98 | 2,56 | 2,19 | 1,86 | 1,57 |
| 4             | 4,72 | 4,12 | 3,59 | 3,10 | 2,67 | 2,28 | 1,94 | 1,64 |
| 5             | 4,89 | 4,28 | 3,72 | 3,22 | 2,77 | 2,37 | 2,02 | 1,70 |
| 6             | 5,08 | 4,44 | 3,87 | 3,35 | 2,88 | 2,47 | 2,10 | 1,77 |
| 7             | 5,27 | 4,61 | 4,01 | 3,48 | 2,99 | 2,56 | 2,18 | 1,85 |
| 8             | 5,46 | 4,78 | 4,17 | 3,61 | 3,11 | 2,67 | 2,27 | 1,92 |
| 9             | 5,65 | 4,96 | 4,32 | 3,75 | 3,23 | 2,77 | 2,36 | 2,00 |
| 10            | 5,86 | 5,14 | 4,49 | 3,89 | 3,36 | 2,88 | 2,45 | 2,08 |
| 11            | 6,06 | 5,32 | 4,65 | 4,03 | 3,48 | 2,98 | 2,54 | 2,16 |
| 12            | 6,28 | 5,52 | 4,82 | 4,18 | 3,61 | 3,10 | 2,64 | 2,24 |
| 13            | 6,49 | 5,71 | 4,99 | 4,33 | 3,74 | 3,21 | 2,74 | 2,33 |
| 14            | 6,72 | 5,91 | 5,17 | 4,49 | 3,88 | 3,34 | 2,85 | 2,42 |
| 15            | 6,93 | 6,11 | 5,35 | 4,65 | 4,02 | 3,46 | 2,95 | 2,51 |
| 16            | 7,17 | 6,32 | 5,54 | 4,82 | 4,17 | 3,59 | 3,07 | 2,60 |
| 17            | 7,40 | 6,53 | 5,72 | 4,99 | 4,32 | 3,71 | 3,18 | 2,70 |

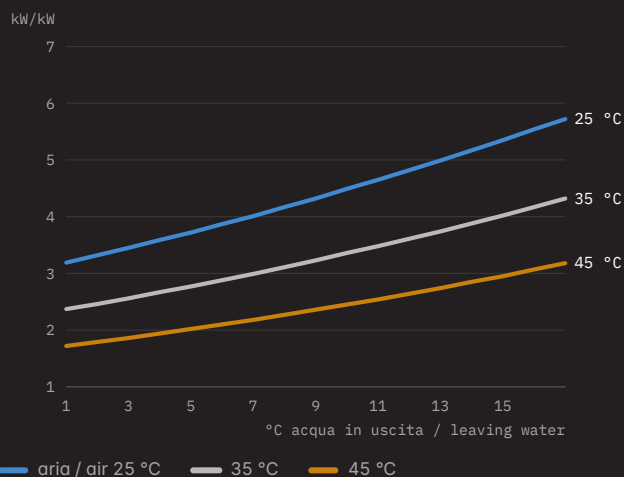
# Curve e dimensioni

## Curves and dimensions

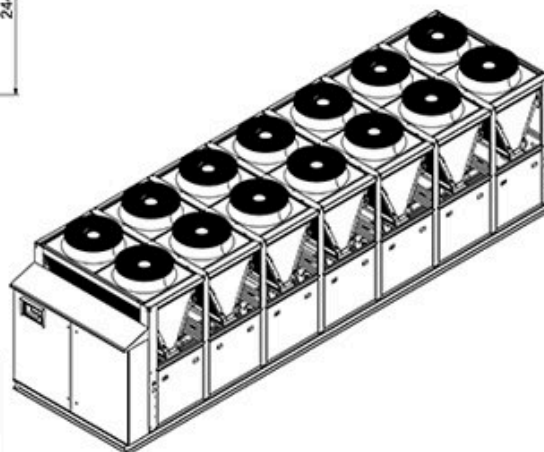
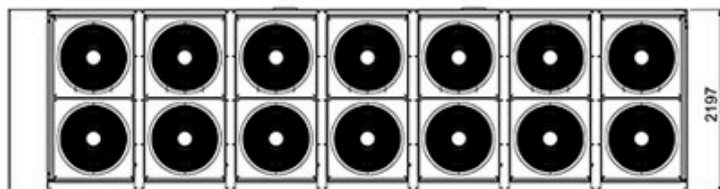
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 8.120 × 2.197 × 2.442 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

PROS9-A950-OSB-M · 66 optional disponibili per questa configurazione · 66 options available for this configuration

## Pompe / Pumps

|                                                                                                    |
|----------------------------------------------------------------------------------------------------|
| Pompa 132 m³/h – prevalenza 25 m<br>Pump 132 m³/h – head 25 m                                      |
| Pompa 132 m³/h – prevalenza 40 m<br>Pump 132 m³/h – head 40 m                                      |
| Pompa doppia montata 132 m³/h – prevalenza 25 m<br>Twin pump, factory-fitted, 132 m³/h – head 25 m |
| Pompa doppia montata 132 m³/h – prevalenza 40 m<br>Twin pump, factory-fitted, 132 m³/h – head 40 m |

## Serbatoi e vasi di espansione / Tanks and expansion vessels

|                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata<br>External water tank 2000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata<br>External water tank 1500 l, AISI 304 with factory-fitted single pump                      |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata<br>External water tank 2000 l, AISI 304 with factory-fitted single pump                      |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory-fitted single pump                      |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory-fitted single pump                      |

|                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 2000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 3000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 5000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1500 l, pressurised, galvanised steel with factory-fitted single pump |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump |
| Vaso di espansione 100 l<br>Expansion vessel 100 l                                                                                                                                         |
| Vaso di espansione 200 l<br>Expansion vessel 200 l                                                                                                                                         |
| Vaso di espansione 300 l<br>Expansion vessel 300 l                                                                                                                                         |
| Vaso di espansione 500 l<br>Expansion vessel 500 l                                                                                                                                         |
| Vaso di espansione 750 l<br>Expansion vessel 750 l                                                                                                                                         |

## Valvole e filtri / Valves and filters

|                                                       |
|-------------------------------------------------------|
| Elettrovalvola 2"<br>Solenoid valve 2"                |
| Elettrovalvola 3"<br>Solenoid valve 3"                |
| Elettrovalvola 4"<br>Solenoid valve 4"                |
| Elettrovalvola 5"<br>Solenoid valve 5"                |
| Elettrovalvola 6"<br>Solenoid valve 6"                |
| Filtro a dischi in acciaio 2"<br>Steel disc filter 2" |

# Optional — 2/2

## Options

PROS9-A950-OSB-M · 66 optional disponibili per questa configurazione · 66 options available for this configuration

### Valvole e filtri / Valves and filters

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

### Free cooling / Free cooling

Free cooling 600 kW – 12 ventilatori (rif. 10)

Free cooling 600 kW – 12 fans (ref. 10)

### Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

**Procryon — Smart Cooling Engineering S.r.l.**  
Via Ugo Vaglia 10 — Gavardo (BS) — Italia  
[www.procryon.com](http://www.procryon.com)



**PROCRYON**

Procryon — Smart Cooling Engineering S.r.l.  
[www.procryon.com](http://www.procryon.com)

Continuity by design